# C13 Halogenoalkanes – Knowledge Organiser

*AQA A-level Chemistry • Set 1: Nucleophilic Substitution* **Paper 2**

| # | Question | Answer |
|---|---|---|
| 1 | Polarity of the C–X bond | $C^{\delta+}–X^{\delta-}$ |
| 2 | Nucleophile | Electron pair donor |
| 3 | Three nucleophiles in the AQA spec | $OH^-$, $CN^-$, $NH_3$ |
| 4 | Trend in C–X bond enthalpy down Group 7 | Decreases |
| 5 | Most reactive halogenoalkane | Iodoalkane |
| 6 | Why iodoalkanes are most reactive | Weakest C–X bond |
| 7 | Reagent and conditions to make an alcohol | NaOH(aq), warm |
| 8 | Reagent and conditions to make a nitrile | KCN in ethanol/water, reflux |
| 9 | Reagent and conditions to make an amine | Excess conc $NH_3$ in ethanol, heat in sealed tube |
| 10 | Product of bromoethane with KCN | Propanenitrile |

# C13 Halogenoalkanes – Knowledge Organiser

*AQA A-level Chemistry • Set 2: Elimination and Classification* **Paper 2**

| # | Question | Answer |
|---|---|---|
| 1 | Reagent and conditions for elimination | KOH in ethanol, hot |
| 2 | Organic product of elimination | Alkene |
| 3 | Role of $OH^-$ in elimination | Base |
| 4 | Role of $OH^-$ in substitution | Nucleophile |
| 5 | Reaction of halogenoalkane with $OH^-(aq)$ | Nucleophilic substitution |
| 6 | Reaction of halogenoalkane with $OH^-$ in ethanol | Elimination |
| 7 | Product of 2-bromopropane with ethanolic KOH | Propene |
| 8 | Primary halogenoalkane | C–X carbon bonded to one other carbon |
| 9 | Secondary halogenoalkane | C–X carbon bonded to two other carbons |
| 10 | Tertiary halogenoalkane | C–X carbon bonded to three other carbons |

# C13 Halogenoalkanes – Knowledge Organiser

*AQA A-level Chemistry • Set 3: Ozone Depletion and Products* **Paper 2**

| # | Question | Answer |
|---|---|---|
| 1 | Formula of ozone | $O_3$ |
| 2 | Role of the ozone layer | Absorbs UV radiation |
| 3 | Compounds that deplete ozone | CFCs |
| 4 | Species that breaks down ozone | Cl• radicals |
| 5 | Overall equation for ozone breakdown | $2O_3 \rightarrow 3O_2$ |
| 6 | Why one Cl• destroys many $O_3$ | Acts as a catalyst (regenerated) |
| 7 | Replacement for CFCs | HFCs |
| 8 | Why HFCs do not deplete ozone | C–F bond too strong to break |
| 9 | Nitrile functional group | –C≡N |
| 10 | Why excess $NH_3$ is used | Reduces further substitution |